
Comparison of Survival Rates for Patients With Uveal Melanoma After Treatment With Proton Beam Irradiation or Enucleation

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We compared survival experiences for three groups with uveal melanoma: (1) 120 patients treated by proton beam irradiation from 1975 to 1981; (2) 235 patients treated by enucleation from 1953 to 1973 (enucleation group 1); and (3) 161 patients treated by enucleation from 1975 to 1981 (enucleation group 2). The following variables were predictive of melanoma-related deaths. Treatment: rate ratio (ratio of the rate of death for one category of the variable relative to the rate of death for another category) of 6.32 (95% confidence interval 1.70 to 23.51) for enucleation group 1 vs proton beam, 3.06 (0.81 to 11.54) for enucleation group 2 vs proton beam, and 2.07 (1.05 to 4.07) for enucleation group 1 vs enucleation group 2. Size: rate ratio of 3.61 (1.51 to 8.64) for large (16+ mm) vs small (<11 mm) tumors. Location: rate ratio of 3.19 (1.56 to 6.53) for tumors anterior vs posterior to the equator. When death from all causes was evaluated as the outcome, the estimated rate ratios were similar. When melanoma metastasis was evaluated as the outcome, the estimated rate ratios were reduced in magnitude.

PROTON BEAM radiation therapy of uveal melanoma is effective in maintaining vision and saving eyes in many cases.¹ The physical characteristics of protons include minimal scatter, maximum dose at the end of the beam path, and minimal irradiation beyond the tumor target. Because of these characteristics, proton

beam irradiation provides highly localized dose distributions that have decreased ocular morbidity and treatment failure associated with some other conservative methods.¹⁻³ However, an important question regarding the use of this modality for the treatment of melanoma remains to be answered. We wondered whether proton beam treatment compares unfavorably to enucleation in terms of the occurrence of metastases and patient survival. We studied this question in a nonrandomized clinical trial, comparing the survival experience of patients treated by proton beam irradiation with that of those treated by enucleation, controlling for some known and potential prognostic factors in the analysis. The outcomes evaluated included the time from treatment to the occurrence of uveal melanoma metastases, melanoma-related deaths, and deaths from all causes.

Subjects and Methods

Subjects—We compared three groups of subjects (Table 1): The proton beam group consisted of the first 120 patients with unilateral uveal melanoma without clinical extrascleral extension treated by proton beam irradiation from July 1975 to December 1981. One patient with extrascleral extension was not included. Details of this technique have been published previously.^{4,5} Enucleation group 1 consisted of 235 patients treated by enucleation from 1953 to 1973 with a diagnosis of uveal melanoma identified by the records of the Eye Pathology Laboratory at the Massachusetts Eye and Ear Infirmary. The latter cases were confirmed by review of histopathologic specimens. These subjects have been described elsewhere⁶ and were part of a group of 267 cases with known survival status out of a total of 431 eligible patients. Thirty-two cases with histologic extrascleral extension were excluded from this sample of 267 cases for the present analysis. Enucleation group 2

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